MECH 325 Machine Elements

Gear Trains

Tutorial 3 – Gear Stress Analysis

1. Adapted from **Shigley, Problem 14-20**. (10th ed 14-19)

   A commercial enclosed gear drive is to be designed. The drive is to consist of two spur gears with a 20° pressure angle. The pinion is to have 16 teeth driving a 48- tooth gear. The pinion speed is 300 rev/min, the face width is 2 in, and the diametral pitch is 6 teeth/in. The gears are Grade 1 steel, through-hardened at 200 Brinell, made to No. 6 quality standards. The gears will be uncrowned, centered on their shafts between bearings, and are to be accurately and rigidly mounted. Assume a pinion life of 10^8 cycles and a reliability of 0.90. The power to be transmitted is 5 hp.

   Part 1: To be demonstrated by the TA.

   a. Determine the AGMA bending stress in the pinion
   b. Determine the corresponding safety factor for the pinion bending stress

   Part 2: To be done by students.

   c. Determine the AGMA contact stress in the pinion
   d. Determine the corresponding safety factor for the pinion contact stress

   Part 3: Extra practice, to be done by students at home.

   e. Repeat the above bending and contact stress analysis for the gear.